Supplement 3

Table 5 Clinical characteristics and treatment outcome in the reviewed case reports and case series

First auther, year of publication (country)	*Age/ Sex*	*Clinical presentations*	*Causes of DV*	*Previous history of other varices*	*Previous history of endoscopic treatments*	*Locations of DV*	*Other varices under endoscpy*	*Endoscopic therapeutic modalities*	*Initial hemostasis*	*Treatment success*	*Adverse events*	***Rebleeding***	*Further treatment*	*DV elimination*	*Death*	*Follow-up (month)*
Chang Lin[1] 2018 (China)	78/M	Melena	LC	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	ND	No	1.0
Hosseini-Carroll, P[2] 2017 (US)	43/F	Hematemesis and melena	PVTT	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	ND	No	ND
Malik, A[3] 2017 (US)	60/F	Hematochezia and melena	LC-A	No	No	Descending part	EV	EBL	Yes	Yes	No	No	No	ND	No	ND
Ehrlich, D[4] 2016 (US)	64/F	Hematochezia	LC-C	ND	ND	Descending part	No	EBL	Yes	Yes	Ulcerative bleeding	Yes	Epinephrine spray	ND	No	1.5
Botros, Y[5] 2015 (US)	59/M	Hematochezia	LC-A	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	ND	No	ND
Chopade, T[6] 2014 (US)	52/M	Hematemesis and melena	Idiopathic HP	EV+GV	EIS	Descending part	No	EBL	Yes	Yes	No	No	No	Yes	No	5.0
Sistla, S. C[7] 2013 (Indian)	25/F	ND	LC	EV	EIS	Bulb part	No	EBL	Yes	Yes	No	No	No	ND	No	18.0
Gunnerson, A. C[8], 2012 (US)	49/M	Hematemesis and hematochezia	LC-A	No	No	ND	EV	EBL	Yes	Yes	No	No	No	No	Yes (MODS)	0.27
Gunnerson, A. C[8], 2012 (US)	50/M	Melena and hematochezia	Vascular abnormali	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	Yes	No	24.0

			ties caused by abdominal surgery												
Gunnerson, A. C[8], 2012 (US)	61/F	Melena	Idiopathic portal hypertension	ND	ND	Descending part	No	EBL	Yes	Yes	No	Yes	EBL	ND	No	21.0
Gunnerson, A. C[8], 2012 (US)	57/M	Hematochezia and melena	LC-A+C	GV	No	Descending and horizontal part	EV	EBL	Yes	Yes	No	No	No	ND	No	9.0
Dulic, M[9], 2011 (Austria)	78/F	Hematemesis and melena	PVTT	No	No	Bulb part	EV	EBL	Yes	Yes	Peptic ulcer	No	No	ND	No	2.0
Schmeltzer, P. A[10], 2011 (US)	41/M	Melena	LC-A	No	No	Descending part	EV	EBL	Yes	Yes	No	No	No	Yes	No	1.0
Trikudanathan, G[11], 2011 (US)	57/M	Hemetemesis and hematochezia	LC-A	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	Yes	No	1.0
Kawakami, Y[12], 2011 (Japan)	72/M	Melena	LC-C	ND	ND	Horizontal part	No	EBL	Yes	Yes	No	No	No	ND	No	ND
Chaaya, A[13] 2010 (US)	53/M	Hematochezia	LC-A+C	ND	ND	Descending part	No	EBL	Yes	No	No	No	No	ND	Yes (Respiratory failure)	ND
Goetz, M[14],	57/N	Hematochezia	CTPV	No	No	Descending	RV	EBL	Yes	Yes	No	No	No	ND	No	4.0

2009 (Germany)	D	and hematemesis				part										
Silva, M[15] 2008 (Brazil)	60/F	ND	LC-C	ND	ND	Bulb part	No	EBL	Yes	Yes	No	No	No	Yes	No	2.0
Tan, N. C[16] 2005 (Singapore)	55/M	Melena	LC-A+B	GV	No	Bulb part	EV	EBL	Yes	Yes	No	No	No	Yes	No	12.0
Fayad, N[17], 2004 (US)	35/F	Hematemesis	LC-A	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	ND	No	ND
Bosch, A[18] 2003 (US)	63/M	Melena	PVT	EV	No	ND	EV	EBL	No	No	No	No	EBL	ND	No	11.0
Al-Khatatneh, Salim[19], 2000 (US)	34/F	Hematemesis and melena	LC-A	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	ND	No	ND
Yeh, Y. Y[20] 1998 (China)	68/M	Hematochezia	HCV-related LC+HCC +PVTT	GV	No	Bulb part	EV+GV	EBL	Yes	Yes	No	No	No	Yes	No	2.0
Seo, Y. S[21] 2008 (Korea)	63/M	Melena	LC+Budd -Chiari syndrome	EV	EIS	Descending part	No	EBL	No	No	No	No	EIS	Yes	No	3.0
Park, Sang Woo[22], 2017 (Korea)	69/F	Hematemesis	LC-B	ND	ND	Descending part	No	EBL	Yes	No	Ulcerative bleeding	No	Hemoclipping	Yes	Yes (HCC)	24.0
Sangha, T[23] 2016 (US)	58/M	Hematochezia	LC-C	EV+GV	No	Descending part	No	EBL	Yes	No	Ulcerative bleeding	No	Hemoclipping+EIS	No	No	1.0

Tiwari, P[24] 2010 (US)	30/M	Melena	LC-A+C	ND	ND	Descending part	No	EBL	Yes	Yes	No	No	No	No	No	ND
Selcuk, H[25] 2004 (Turkey)	38/M	Hematemesis	HBV-related LC+Budd-Chiari syndrome	No	No	Horizontal part	EV	EBL	Yes	Yes	No	No	No	No	No	1.0
Akazawa, Y[26], 2003 (Japan)	62/M	Melena	LC-C	ND	ND	Descending part	No	EBL	Yes	Yes	Peptic ulcer	No	No	No	No	0.77
Yoshida, Y[27] 1997 (Japan)	63/M	Melena	LC-A	EV	EBL	Descending part	EV	EBL	Yes	Yes	Peptic ulcer	Yes	ECA	No	No	0.23
Shiraishi, M[28], 1999 (Japan)	56/F	Melena	LC-C	No	No	Bulb and descending part	EV+GV	EIS	No	No	No	Yes	EBL	No	No	0.75
Temel, T[29] 2016 (Turkey)	48/F	ND	LC	No	No	Bulb part	EV	EIS	Yes	Yes	No	No	No	Yes	No	12.0
Steevens, C[30] 2015 (US)	61/M	Hematochezia and melena	Vascular abnormality caused by abdominal surgery	ND	ND	Descending and horizontal part	No	EIS	Yes	Yes	No	No	No	Yes	No	20.0
Steevens, C[31] 2013 (US)	64/M	Melena	Vascular abnormality caused by	No	No	Descending part	CV	EIS	Yes	Yes	No	No	No	Yes	No	1.0

			abdominal surgery													
Culliford, Andrea[32], 2009 (US)	54/M	Hematochezia	LC-A+C	ND	ND	ND	No	EIS	Yes	No	No	No	No	No	Yes (Bleeding)	0.07
Machida, T[33] 2006 (Japan)	79/F	Hematemesis and melena	LC	GV	EBL+EIS	Descending part	No	EIS	No	No	No	No	No	No	Yes (Bleeding)	0.03
Hashiguchi, M[34], 1999 (Japan)	48/M	ND	LC	ND	ND	Descending part	No	EIS	Yes	Yes	No	Yes	No	ND	Yes (Bleeding)	ND
Khan, M. Q[35], 1999 (Saudi Arabia)	45/M	Hematochezia	LC-A	No	No	Horizontal part	EV	EIS	Yes	Yes	Peptic ulcer	No	No	Yes	No	24.0
Jonnalagadda, S. S[36], 1998 (US)	48/M	Melena	LV-A+B+C	EV	EIS	Descending part	No	EIS	No	No	No	No	TIPS	No	No	0.03
Wu, C. S[37] 1995 (China)	43/M	Melena	LC-A	EV	No	Descending part	EV	EIS	Yes	Yes	No	No	No	No	No	1.0
Herold, G[38] 1993 (Germany)	37/ND	ND	ND	EV+GV	No	Bulb part	No	EIS	Yes	Yes	No	No	No	ND	No	10.0
Barbish, A. W[39], 1993 (US)	50/M	HE	LC-A	EV	EIS	Bulb and descending part	EV+GV	EIS	Yes	Yes	Abdominal pain	No	No	ND	No	12.0
Barlas, S[40] 1992 (Saudi	35/M	Melena	Inferior vena cava	ND	ND	Descending part	No	EIS	Yes	Yes	No	No	No	Yes	No	6.0

Arabia)			thrombophlebitis with thrombosis													
Nardone, G[41] 1991 (Italy)	75/F	Melena	Cryptogenetic LC	EV	EIS	Bulb and descending part	No	EIS	Yes	Yes	No	No	No	Yes	No	2.0
Tsuji, H[42] 1989 (Japan)	38/M	Melena	Idiopathic LC	EV	EIS	Descending part	No	EIS	Yes	Yes	Peptic ulcer	No	No	ND	No	14.0
Gertsch, P[43] 1988 (Switzerland)	40/F	ND	LC-A	EV	EIS	Descending part	EV	EIS	Yes	Yes	No	Yes	EIS	Yes	No	36.0
Kirkpatrick, J. R[44], 1985 (Canada)	36/M	Melena	LC-A	EV	EIS	Descending part	EV	EIS	Yes	Yes	Peptic ulcer	No	No	No	No	0.33
Sauerbruch, T[45], 1982 (Germany)	42/M	Hematemesis and melena	Cryptogenetic LC	EV	EIS	Bulb part	EV+GV	EIS	Yes	Yes	Peptic ulcer	No	No	ND	No	18.0
Campo, R[46], 1996 (Spain)	63/ND	Hematemesis	LC	ND	ND	Descending part	No	EIS	No	No	No	No	Surgery (duodenal resection)	No	Yes (MODS)	0.13
Sans, M[47], 1996 (Spain)	22/M	ND	LC	EV+GV	EIS	Bulb part	No	EIS	Yes	Yes	Peptic ulcer	No	No	ND	No	5.0
Khouqeer, F[48], 1987 (US)	37/M	ND	LC-A	No	No	Descending part	EV	EIS	No	No	No	No	Surgery (duodenal	No	Yes (liver	0.03

													resection)		failure)	
Sonpal, N[49], 2015 (US)	55/M	Hematemesis	LC-A	No	No	Descending part	EV	EIS	No	No	No	No	ECA	No	No	0.03
Wu, W. C[50] 2002 (China)	58/F	Hematemesis	LC-C	EV+GV	EBL	Descending part	EV+GV	ETA	Yes	Yes	Pulmonary embolism	No	No	ND	No	ND
Benedetti, G[51], 1993 (Italy)	43/F	Hematemesis	LC-A	No	No	Descending part	EV	ETA	Yes	Yes	Pulmonary embolism	No	No	Yes	No	8.0
Park, Sang Woo[22], 2017 (Korea)	72/M	Hematemesis	LC-B	ND	ND	Descending part	No	ETA	Yes	Yes	No	Yes	ECA	ND	Yes (liver failure)	30.0
Park, Sang Woo[22], 2017 (Korea)	55/M	Melena	LC-A	No	No	Descending and horizontal part	No	ETA	Yes	Yes	Portal vein thrombosis	No	No	Yes	No	14.0
Rana, S. S[52] 2014 (Indian)	35/M	ND	EHPVO	EV	ECA	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	4.0
Rana, S. S[52] 2014 (Indian)	27/M	ND	EHPVO	ND	ND	Bulb part	No	ETA	Yes	Yes	No	No	No	No	No	ND
Rana, S. S[52] 2014 (Indian)	38/M	ND	LC-A	EV	ECA	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	ND
Rana, S. S[52] 2014 (Indian)	28/M	ND	EHPVO	EV	ECA	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	46.0
Kakizaki, S[53] 2010 (Japan)	47/F	Melena	LC-B	ND	ND	Descending part	No	ETA	Yes	Yes	No	No	DBOE	No	No	0.27
Sousa, H. T[54] 2008	47/F	Hematochezia	LC-A	EV	No	Descending and horizontal	EV	ETA	No	No	No	No	EBL	No	No	0.03

(Portugal)						part									
Illuminati, G[55], 2000 (France)	40/M	Melena	LC-A	No	No	Descending and horizontal part	EV	ETA	Yes	No	No	No	TIPS	No	No	0.07
Abbar, B[56] 2018 (France)	52/M	Hematemesis and melena	LC-B	No	No	Descending part	EV	ETA	Yes	Yes	Cerebral infarction	No	No	ND	No	3.0
Matsui, S[57] 2018 (Japan)	69/F	Hematemesis	ND	ND	ND	Descending part	No	ETA	Yes	No	No	No	No	ND	No	ND
Bhagani, S[58] 2017 (UK)	57/M	Melena	LC-PBC	No	No	Descending part	EV+GV	ETA	Yes	Yes	No	No	No	Yes	No	4.0
Zubaidah, N. H[59], 2016 (Malaysia)	61/F	Melena	Idiopathic LC	ND	ND	Horizontal part	No	ETA	Yes	Yes	No	No	No	Yes	No	ND
Tulio, M. A[60] 2017 (Portugal)	56/M	Melena	LC-C	No	No	Descending part	EV	ETA	Yes	Yes	Peptic ulcer	No	No	Yes	No	24.0
Katz-Agranov, N. S[61], 2016 (US)	46/F	ND	LC-A	No	No	Horizontal part	EV	ETA	Yes	Yes	Ulcerative bleeding	Yes	TIPS	Yes	No	2.0
Rew, J. S[62] 2016 (Korea)	55/M	Melena	LC-A	ND	ND	Descending and horizontal part	No	ETA	Yes	Yes	Biliary obstruction	No	ERCP	Yes	No	6.0
Costa, M[63] 2015 (Portugal)	48/M	Hematemesis and melena	LC-C	No	No	Descending part	EV	ETA	Yes	Yes	No	No	No	Yes	No	6.0
Khan, A. N[64]	35/M	Hematemesis	PVT	No	No	Bulb part	EV	ETA	Yes	Yes	No	No	No	No	No	1.0

2014 (Pakistan)		and melena														
Sharma, B[65] 2014 (Indian)	32/M	Hematemesis and melena	PVT	ND	ND	Bulb part	No	ETA	Yes	Yes	No	No	No	No	No	ND
Lakhtakia, S[66], 2014 (Indian)	54/M	Hematemesis	PVT	ND	ND	Descending part	No	ETA	Yes	Yes	Biliary obstruction	No	ERCP	ND	No	5.0
Salgueiro, P[67], 2014 (Portugal)	45/ND	Melena	LC-C	ND	ND	Descending part	No	ETA	Yes	Yes	No	No	No	ND	No	7.0
Mora-Soler, A[68], 2013 (Spain)	55/F	Hematemesis and melena	CTPV	GV	EBL	Bulb part	EV	ETA	Yes	No	No	No	No	No	Yes (Bleeding)	0.07
Bhat, G[69] 2013 (Indian)	24/M	Hematemesis and melena	CTPV	EV+GV	EBL	Descending part	EV	ETA	Yes	Yes	Biliary obstruction	No	No	ND	No	24.0
Kim, H. H[70] 2012 (Korea)	65/M	Melena	LC-B	ND	ND	Horizontal part	EV	ETA	Yes	Yes	No	No	No	Yes	No	24
Kao, W. Y[71] 2012 (China)	73/M	Hematochezia	CTPV	EV	No	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	2.0
Kang, H. Y[72] 2011 (Korea)	62/M	Hematemesis and melena	LC-A	No	No	Descending and horizontal part	EV	ETA	Yes	Yes	Peptic ulcer	No	No	Yes	No	9.0
Weismuller, T. J[73], 2011 (Germany)	71/M	Melena	Vascular abnormality caused	ND	ND	Descending part	No	ETA	Yes	Yes	Portal vein thrombosis	No	No	Yes	No	9.0

			by abdominal surgery													
Malik, A[74] 2011 (UK)	52/M	Hematochezia	LC-A	ND	ND	Descending part	No	ETA	Yes	Yes	No	No	No	ND	No	ND
Takamatsu, Toru[75], 2011 (Japan)	73/F	Melena	LC	No	No	Descending part	EV	ETA	Yes	Yes	Peptic ulcer	No	No	ND	No	3.0
Ryan, L[76] 2010 (US)	54/F	Hematemesis and melena	LC-A	No	No	Descending part	EV	ETA	No	No	No	No	TIPS	No	No	0.03
Wahl, P[77] 2004 (Switzerland)	57/M	Hematochezia	LC-A	EV	EBL	ND	EV	ETA	Yes	Yes	Sepsis	No	No	ND	No	12.0
Bhasin, D. K[78], 1999 (Indian)	50/F	Hematemesis and melena	PVT	GV	EIS	Bulb part	RV	ETA	Yes	Yes	No	No	No	ND	No	12.0
Ota, K[79] 1998 (Japan)	75/F	Melena	LC-C	No	No	Horizontal part	EV	ETA	Yes	Yes	Pneumonia	No	No	Yes	Yes (Liver failure)	1.33
Paterlini, A[80] 1993 (Italy)	77/M	Melena	LC	No	No	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	18.0
Sung, J. Y[81] 1993 (China)	65/M	ND	PVTT	ND	ND	Bulb part	No	ETA	Yes	Yes	No	No	No	No	No	1.0
Sung, J. Y[81] 1993 (China)	42/M	Hematemesis and melena	LC+HCC +PVTT	EV	ECA	Bulb part	EV	ETA	Yes	Yes	No	No	No	ND	No	ND
Sato, H[82]	64/F	Melena	LC-C	No	No	Horizontal part	EV	EIS+ETA	Yes	Yes	No	No	No	Yes	No	48.0

2002 (Japan)																
Suzuki, R[83] 2009 (Japan)	50/M	Hematemesis	LC-A	EV	EBL	Horizontal part	No	EIS+ ETA	Yes	Yes	No	No	No	Yes	No	13.0
Hoshi, S[84] 2001 (Japan)	57/M	Hematemesis and melena	LC-C	No	No	Bulb part	EV	EIS+ETA	Yes	Yes	No	No	No	Yes	No	2.0
Omata, F[85] 1998 (Japan)	69/F	Melena	LC	No	No	Descending part	EV+GV	EIS+EBL	Yes	Yes	No	No	No	Yes	No	3.0
Mora-Soler, A[68], 2013 (Spain)	54/M	Hematemesis and melena	LC-A	GV	EBL	Descending part	EV	EIS+ ETA	Yes	Yes	No	No	No	ND	No	13.0
Mora-Soler, A[68], 2013 (Spain)	69/M	HE	LC-A	ND	ND	Descending part	No	EIS+ ETA	Yes	No	No	No	No	No	Yes (Liver failure)	0.13
Mora-Soler, A[68], 2013 (Spain)	65/F	Hematochezia	LC-A	ND	ND	Descending part	No	EIS+EBL	No	No	No	No	ECA	No	Yes (Respiratory failure)	3.0
Mora-Soler, A[68], 2013 (Spain)	33/M	Melena	LC-A	No	No	Bulb part	EV+GV	EIS+ ETA	Yes	Yes	No	No	No	ND	No	5.0
Chandrasekar, V. T[86], 2013 (Indian)	35/M	Hematemesis and melena	LC-A	No	No	Descending part	EV	EIS+EBL	Yes	Yes	No	No	No	No	No	0.17
Miyakoda, K[87], 2012 (Japan)	76/F	Melena	LC-C	ND	ND	Descending part	No	EIS+ ETA	Yes	Yes	Embolism in the atrium	No	No	No	Yes (Liver failure)	1.0
Labenz, J[88]	50/N	ND	LC-A	GV	EBL	Descending	EV	EIS+ ETA	Yes	Yes	Peptic	Yes	EIS+ECA	ND	No	12.0

1993 (Germany)	D					part					ulcer				

ND，not described; DV, duodenal varices; EV, esophageal varices; GV, gastric varices; RV, rectal varices; CV, colonic varices; M, male; F, female; LC-A = liver cirrhosis duo to alcoholic liver disease; LC-B, liver cirrhosis due to hepatitis B viral infection; LC-C, liver cirrhosis due to hepatitis C viral infection; LC-A+B, liver cirrhosis due to alcoholic liver disease and hepatitis B viral infection; LC-A+C, liver cirrhosis due to alcoholic liver disease and hepatitis C viral infection; LC-A+B+C, liver cirrhosis due to alcoholic liver disease, hepatitis B viral infection and hepatitis C viral infection; PBC, primary biliary cirrhosis; HCC，hepatocellular carcinoma; PVT, portal vein thrombosis; PVTT, Portal vein tumor thrombosis; EHPVO, extrahepatic portal vein occlusion ; CTPV, Cavernous transformation of portal vein; PH, portal hypertension; HE, Hepatic encephalopathy; MODS, multiple organ dysfunction syndrome; TIPS, transjugular intrahepatic portosystemic shunt; DBOE, Double balloon-occluded embolotherapy; ERCP, Endoscopic retrograde cholangiopancreatography; EBL, endoscopic band ligation; EIS, endoscopic injection sclerotherapy; ETA, endoscopic tissue adhesive; Combination, EIS combined with EBL or ECA.

Reference

1 Lin IC, CT Lee, YC Hsu, and CM Tseng. An unusual cause of upper gastrointestinal bleeding. Dig Liver Dis 2018;50:92.

2 Hosseini-Carroll P and A Seth. GI bleeding: The unusual suspect. Gastrointestinal Endoscopy 2017;85:AB148.

3 Malik A, A Aleem, E Nellis, and H Shah. Duodenal varices: A rare cause of gastrointestinal bleeding. American Journal of Gastroenterology 2017;112:S1032-S1034.

4 Ehrlich D, AM Thaker, D Eshtiaghpour, and M Kaneshiro. Severe bleeding from duodenal varices in a post-liver transplant patient with recurrent cirrhosis. American Journal of Gastroenterology 2016;111:S925.

5 Botros Y, F Asslo, J Shami, and W Baddoura. Endoscopic variceal banding as an initial approach to a bleeding duodenal varix. American Journal of Gastroenterology 2015;110:S205-S206.

6 Chopade T, D Halegoua-De Marzio, A Siddiqui, J Civan, and J Fenkel. Treatment of duodenal variceal hemorrhage with endoscopic band ligation. American Journal of Gastroenterology 2014;109:S367.

7 Sistla SC, VS Karthikeyan, D Ram, SM Ali, and N Rajkumar. Endoscopic variceal ligation as a safe and effective alternative to endoscopic sclerotherapy in bleeding ectopic duodenal varix--report of a rare case. Indian J Gastroenterol 2013;32:419-20.

8 Gunnerson AC, DL Diehl, VN Nguyen, MJ Shellenberger, and J Blansfield. Endoscopic duodenal variceal ligation: a series of 4 cases and review of the literature (with video). Gastrointest Endosc 2012;76:900-4.

9 Dulic M, E Dulic-Lakovic, B Hellmich, B Blaha, and M Gschwantler. Successful treatment of a bleeding duodenal varix by endoscopic band ligation. Journal of Gastrointestinal and Liver Diseases 2011;20:234.

10 Schmeltzer PA and MT Smith. Duodenal variceal bleeding successfully treated with endoscopic banding (with video). Gastrointest Endosc 2011;74:716-7.

11 Trikudanathan G, M Myers, M Karasik, and JS Weiser. Gastrointestinal: Successful obliteration of bleeding duodenal varices using band ligation. Journal of Gastroenterology and Hepatology 2011;26:209-209.

12 Kawakami Y, H Takahashi, S Kato, T Uchiyama, H Mawatari, H Iida, et al. Effective treatment for duodenal variceal bleeding using endoscopic ligation. Hepato-Gastroenterology 2011;58:2024-2025.

13 Chaaya A, L Salese, S Peikin, and L Chaptini. Portal hypertension presenting as isolated duodenal variceal bleed. American Journal of Gastroenterology 2010;105:S370-S371.

14 Goetz M, FK Rahman, PR Galle, and R Kiesslich. Duodenal and rectal varices as a source of severe upper and lower gastrointestinal bleeding. Endoscopy 2009;41 Suppl 2:E169.

15 Silva M, A Mattos, J Wilney, A Cassal, J Bortoluzzi, and P Fontes. Ruptured duodenal varices: The role of banding ligation. American Journal of Case Reports 2008;9:43-46.

16 Tan NC, S Ibrahim, and KH Tay. Successful management of a bleeding duodenal varix by endoscopic banding. SMJ Singapore Medica1 Journal 2005;46:723-725.

17 Fayad N, F Nammour, and A Elfant. Endoscopic variceal ligation for bleeding duodenal varices. J Clin Gastroenterol 2004;38:467.

18 Bosch A, L Marsano, and GW Varilek. Successful obliteration of duodenal varices after endoscopic ligation. Dig Dis Sci 2003;48:1809-12.

19 Al-Khatatneh S, J Fares, H Elfarra, and W Baddoura. Endoscopic band ligation of bleeding duodenal varices. American Journal of Gastroenterology 2000;95:2618-2618.

20 Yeh YY, MC Hou, HC Lin, FY Chang, and SD Lee. Case report: Successful obliteration of a bleeding duodenal varix using endoscopic ligation. Journal of Gastroenterology and Hepatology 1998;13:591-593.

21 Seo YS, YD Kwon, S Park, B Keum, BJ Park, YS Kim, et al. Complete eradication of duodenal varices after endoscopic injection sclerotherapy with ethanolamine oleate: a case report. Gastrointest Endosc 2008;67:759-62.

22 Park SW, E Cho, CH Jun, SK Choi, HS Kim, CH Park, et al. Upper gastrointestinal ectopic variceal bleeding treated with various endoscopic modalities Case reports and literature review. Medicine 2017;96.

23 Sangha T, J Yang, and W Harford. Ectopic varices presenting as a persistent gastrointestinal bleed. American Journal of Gastroenterology 2016;111:S1371.

24 Tiwari P, A Khan, and S Connolly. Ectopic variceal bleed managed conservatively. A clinical perspective. American Journal of Gastroenterology 2010;105:S377.

25 Selcuk H, F Boyvat, S Eren, M Korkmaz, G Gur, U Yilmaz, et al. Duodenal varices as an unusual cause of gastrointestinal bleeding due to portal hypertension: a case report. Turk J Gastroenterol 2004;15:104-7.

26 Akazawa Y, I Murata, T Yamao, M Yamakawa, Y Kawano, N Nomura, et al. Successful management of bleeding duodenal varices by endoscopic variceal ligation and balloon-occluded retrograde transvenous obliteration. Gastrointest Endosc 2003;58:794-7.

27 Yoshida Y, Y Imai, M Nishikawa, M Nakatukasa, M Kurokawa, K Shibata, et al. Successful endoscopic injection sclerotherapy with N-butyl-2-cyanoacrylate following the recurrence of bleeding soon after endoscopic ligation for ruptured duodenal varices. Am J Gastroenterol 1997;92:1227-9.

28 Shiraishi M, S Hiroyasu, T Higa, S Oshiro, and Y Muto. Successful management of ruptured duodenal varices by means of endoscopic variceal ligation: report of a case. Gastrointest Endosc 1999;49:255-7.

29 Temel T, A Aktas, SM Ozgenel, and A Ozakyol. Complete Eradication of Bleeding Duodenal Varices with Endoscopic Polydocanol Sclerotherapy. Euroasian J Hepatogastroenterol 2016;6:176-178.

30 Steevens C, M Abdalla, TH Kothari, V Kaul, and S Kothari. Massive duodenal variceal bleed; complication of extra hepatic portal hypertension: Endoscopic management and literature review. World J Gastrointest Pharmacol Ther 2015;6:248-52.

31 Steevens C, M Abdalla, V Kaul, and S Kothari. Sclerotherapy for duodenal varices associated with mesenteric vein stenosis. American Journal of Gastroenterology 2013;108:S291.

32 Culliford A, J Kurz, I Suleiman, V Gundareddy, and G Mujtaba. Duodenal Varices: A Case Report. American Journal of Gastroenterology 2009;104:S283-S283.

33 Machida T, K Sato, A Kojima, J Takezawa, N Sohara, S Kakizaki, et al. Ruptured duodenal varices after endoscopic ligation of esophageal varices: an autopsy case. Gastrointestinal Endoscopy 2006;63:352-354.

34 Hashiguchi M, H Tsuji, J Shimono, K Azuma, and M Fujishima. Ruptured duodenal varices: an autopsy case report. Hepatogastroenterology 1999;46:1751-4.

35 Khan MQ, S Al-Momen, and A Alghssab. Duodenal varices causing massive lower gastrointestinal hemorrhage. Annals of Saudi Medicine 1999;19:440-443.

36 Jonnalagadda SS, S Quiason, and OJ Smith. Successful therapy of bleeding duodenal varices by TIPS after failure of sclerotherapy. Am J Gastroenterol 1998;93:272-4.

37 Wu CS, CM Chen, and KY Chang. Endoscopic injection sclerotherapy of bleeding duodenal varices. J Gastroenterol Hepatol 1995;10:481-3.

38 Herold G and EF Stange. Sclerotherapy of duodenal varices using a fibrin tissue sealant. Endoscopy 1993;25:371-2.

39 Barbish AW and MN Ehrinpreis. Successful endoscopic injection sclerotherapy of a bleeding duodenal varix. Am J Gastroenterol 1993;88:90-2.

40 Barlas S, S Chaudhry, and M Al-Rashid. Bleeding duodenal varices in a patient with Behcet's disease: A case report. Ann Saudi Med 1992;12:212-4.

41 Nardone G and G Budillon. Treatment of duodenal varices by endoscopic sclerotherapy. Gastrointest Endosc 1991;37:407-8.

42 Tsuji H, H Okano, H Fujino, T Satoh, T Kodama, T Takino, et al. A case of endoscopic injection sclerotherapy for a bleeding duodenal varix. Gastroenterologia Japonica 1989;24:60-4.

43 Gertsch P and LH Blumgart. CURE OF A BLEEDING DUODENAL VARIX BY SCLEROTHERAPY. British Journal of Surgery 1988;75:717-717.

44 Kirkpatrick JR, JP Shoenut, and AB Micflikier. Successful injection sclerotherapy for bleeding duodenal varix in intrahepatic portal obstruction. Gastrointest Endosc 1985;31:259-60.

45 Sauerbruch T, M Weinzierl, HP Dietrich, G Antes, J Eisenburg, and G Paumgartner. SCLEROTHERAPY OF A BLEEDING DUODENAL VARIX. Endoscopy 1982;14:187-189.

46 Campo R, E Brullet, A Corcuera, and D Joseph. Vanishing duodenal varix: unreported presentation causing massive bleeding. Endoscopy 1996;28:789.

47 Sans M, J Llach, JM Bordas, V Andreu, JC Reverter, J Bosch, et al. Thrombin and ethanolamine injection therapy in arresting uncontrolled bleeding from duodenal varices. Endoscopy 1996;28:403.

48 Khouqeer F, C Morrow, and P Jordan. Duodenal varices as a cause of massive upper gastrointestinal bleeding. Surgery 1987;102:548-52.

49 Sonpal N, P Mudireddy, A Mammen, and G Haber. Cyanoacrylate to the rescue: Successful treatment of duodenal variceal bleeding by glue injection. American Journal of Gastroenterology 2015;110:S454.

50 Wu WC, LY Wang, FJ Yu, WM Wang, SC Chen, WL Chuang, et al. Bleeding duodenal varices after gastroesophageal varices ligation: a case report. Kaohsiung J Med Sci 2002;18:578-81.

51 Benedetti G, R Sablich, T Lacchin, and A Masiero. Endoscopic treatment of bleeding duodenal varices by bucrylate injection. Endoscopy 1993;25:432-3.

52 Rana SS, DK Bhasin, V Sharma, V Chaudhary, R Sharma, and K Singh. Clinical, endoscopic and endoscopic ultrasound features of duodenal varices: A report of 10 cases. Endoscopic Ultrasound 2014;3:54-57.

53 Kakizaki S, M Toyoda, T Ichikawa, K Sato, H Takagi, H Arai, et al. Clinical characteristics and treatment for patients presenting with bleeding duodenal varices. Digestive Endoscopy 2010;22:275-281.

54 Sousa HT, C Gregorio, P Amaro, M Ferreira, JM Romaozinho, H Gouveia, et al. Successful endoscopic banding after cyanoacrylate failure for active bleeding duodenal varix. Revista Espanola De Enfermedades Digestivas 2008;100:171-172.

55 Illuminati G, A Smail, D Azoulay, D Castaing, and H Bismuth. Association of transjugular intrahepatic portosystemic shunt with embolization in the treatment of bleeding duodenal varix refractory to sclerotherapy. Dig Surg 2000;17:398-400.

56 Abbar B, D Vimpere, S Khater, C Cellier, and JL Diehl. Cerebral infarction following cyanoacrylate endoscopic therapy of duodenal varices in a patient with a patent foramen ovale. Intern Emerg Med 2018;13:817-819.

57 Matsui S, H Kashida, and M Kudo. Utility of endoscopic ultrasound in hemorrhage from recurrent duodenal varices. Annals of Gastroenterology 2018;31:636.

58 Bhagani S, C Winters, and S Moreea. Duodenal variceal bleed: an unusual cause of upper gastrointestinal bleed and a difficult diagnosis to make. BMJ Case Rep 2017;2017.

59 Zubaidah NH, JH Asraf, L Lee, and T Gee. Idiopathic duodenal varix presenting as a massive upper gastrointestinal bleeding: A case report. Med J Malaysia 2016;71:294-295.

60 Tulio MA, S Marques, M Bispo, T Bana, and C Chagas. Endoscopic Management of Acute Bleeding from an Ectopic Duodenal Varix. GE Port J Gastroenterol 2017;24:98-100.

61 Katz-Agranov NS, MI Nevah Rubin, and MB Fallon. Treating duodenal varices with cyanoacrylate glue injection. American Journal of Gastroenterology 2016;111:S695.

62 Rew JS, CH Jun, and E Cho. A rare case of portal biliopathy and pylephlebitis following the injection of cyanoacrylate into the duodenal varix. Journal of Gastroenterology and Hepatology (Australia) 2016;31:295.

63 Costa M and G Ramos. Duodenal Variceal Bleeding Successfully Treated by Endoscopic N-butyl-2-cyanoacrylate Injection. GE Port J Gastroenterol 2015;22:180-181.

64 Khan AN, IG Kiani, M Arshad, R Hidayat, K Said, and A Shehzad. Management of ectopic varix with histoacryl. J Ayub Med Coll Abbottabad 2014;26:618-20.

65 Sharma B, S Raina, and R Sharma. Bleeding ectopic varices as the first manifestation of portal hypertension. Case Reports Hepatol 2014;2014:140959.

66 Lakhtakia S, BR Pericherla, KV Bhardwaj, MJ Ramchandani, R Gupta, and ND Reddy. Common bile duct compression caused by cyanoacrylate glue injection into bleeding duodenal varices: a rare adverse event (with videos). Gastrointest Endosc 2014;79:1005-6: discussion 1007.

67 Salgueiro P, T Araújo, T Moreira, P Lago, and I Pedroto. Endoscopic injection of a ruptured duodenal varix with n-butyl-2-cyanoacrylate. Video Journal and Encyclopedia of GI Endoscopy 2014;2:26-28.

68 Mora-Soler A, A Velasco-Guardado, R Acosta-Materan, J Umana-Mejia, Y Jamanca-Poma, R Calderon-Begazo, et al. Endoscopic treatment of duodenal varices with cyanoacrylate. Rev Esp Enferm Dig 2013;105:629-32.

69 Bhat G, D Suvarna, and CG Pai. Symptomatic portal biliopathy following cyanoacrylate injection for a bleeding duodenal varix. Trop Gastroenterol 2013;34:108-9.

70 Kim HH and SE Kim. Ruptured Duodenal Varices Successfully Managed by Endoscopic N-butyl-2-cyanoacrylate Injection. J Clin Med Res 2012;4:351-3.

71 Kao WY, WC Wu, PH Chen, and YY Chiou. Duodenal variceal bleeding caused by chronic pancreatitis. Gastrointest Endosc 2012;75:922-3.

72 Kang HY, WK Lee, YH Kim, BW Kwon, MS Kang, SB Kim, et al. Ruptured duodenal varices arising from the main portal vein successfully treated with endoscopic injection sclerotherapy: a case report. Korean J Hepatol 2011;17:152-6.

73 Weismuller TJ, N Schweitzer, B Boozari, A Negm, D Burger, MP Manns, et al. Peripapillary duodenal varices as a rare cause of severe bleeding in a patient with no other signs of portal hypertension--successful endoscopic treatment with cyanoacrylate injection. Z Gastroenterol 2011;49:591-5.

74 Malik A, N Junglee, A Khan, J Sutton, J Gasem, and W Ahmed. Duodenal varices successfully treated with cyanoacrylate injection therapy. BMJ Case Rep 2011;2011.

75 Takamatsu T, H Ootake, T Uehara, Y Shindou, T Ikeya, K Toukai, et al. A case of ruptured duodenal varices treated successfully by endoscopic injection sclerotherapy under radiographic guidance with a mixture of N-butyl-2-cyanoacrylate-lipiodol. Jichi Med Univ J 2011;34:87-95.

76 Ryan L, R Bethune, and K Habeeb. Haematemesis and melaena [corrected] as the first presentation of duodenal varices. J R Soc Med 2010;103:109-11.

77 Wahl P, F Lammer, D Conen, R Schlumpf, and A Bock. Septic complications after injection of N-butyl-2-cyanoacrylate: report of two cases and review. Gastrointestinal Endoscopy 2004;59:911-916.

78 Bhasin DK, BC Sharma, PV Sriram, G Makharia, and K Singh. Endoscopic management of bleeding ectopic varices with histoacryl. HPB Surg 1999;11:171-3.

79 Ota K, Z Shirai, T Masuzaki, K Tanaka, H Higashihara, M Okazaki, et al. Endoscopic injection sclerotherapy with n-butyl-2-cyanoacrylate for ruptured duodenal varices. J Gastroenterol 1998;33:550-5.

80 Paterlini A, F Rolfi, F Buffoli, P Cesari, M Graffeo, D Benedini, et al. Endoscopic treatment of a bleeding duodenal varix using N-Butyl-2-Cyanoacrylate [4]. Endoscopy 1993;25:434.

81 Sung JY, SC Chung, HT Leung, CW Lai, AE Ismael, and AK Li. Duodenal varices in hepatocellular carcinoma. Endoscopy 1993;25:194-6.

82 Sato H, H Fujishiro, MA Rumi, Y Kinoshita, M Niigaki, N Kohge, et al. Successful endoscopic injection sclerotherapy for duodenal varices. J Gastroenterol 2002;37:143-4.

83 Suzuki R, A Irisawa, T Hikichi, G Shibukawa, T Takagi, T Wakatsuki, et al. Hemorrhagic duodenal varices treated successfully with endoscopic injection sclerotherapy using cyanoacrylate and ethanolamine-oleate: a case report. Surg Laparosc Endosc Percutan Tech 2009;19:e233-6.

84 Hoshi S. Ruptured duodenal varix. Gastrointest Endosc 2001;53:86.

85 Omata F, T Itoh, Y Shibayama, H Ide, H Takahashi, F Ueno, et al. Duodenal variceal bleeding treated with a combination of endoscopic ligation and sclerotherapy. Endoscopy 1998;30:S62-3.

86 Chandrasekar VT, GB Janakan, R Venkatachalam, and CT Seshadri. Treatment and follow-up of a case of bleeding duodenal varix. Journal of Digestive Endoscopy 2013;4:128-130.

87 Miyakoda K, H Takedatsu, K Emori, H Inoue, A Toyonaga, K Mitsuyama, et al. N-butyl-2-cyanoacrylate (histoacryl) glue in the right atrium after endoscopic injection for a ruptured duodenal varix: complication of histoacryl injection. Dig Endosc 2012;24:192.

88 Labenz J and G Borsch. Successful endoscopic hemostasis of duodenal variceal bleeding with histoacryl. Endoscopy 1993;25:194.